

MEDICAL INSURANCE PRICE PREDICTION

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

BY

SHAIK MOHAMMAD HUSSAIN (23501A05F5)

ROWTHU DEVIKA (23501A05F1)

RAMAVATH SHIVA RAM (23501A05E8)

PATNALA CHANDRA VENKATA NAGA RAJU (23501A05D9)

RUTHVIK SRI SAI CHAGARLAMUDI (23501A05F2)



PRASAD V POTLURI SIDDHARTHA INSTITUTE OF TECHNOLOGY

(Permanently affiliated to JNTU Kakinada, Approved by AICTE)

(An NBA & NAAC A+ accredited and ISO 21001:2018 Certified Institution)

Kanuru, Vijayawada – 520007

(2025-26)

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CERTIFICATE

This is to certify that the project report title “**Medical Insurance Price Prediction**” is the bonafide work of **SHAIK MOHAMMAD HUSSAIN (23501A05F5)**, **ROWTHU DEVIKA (23501A05F1)**, **RAMAVATH SHIVA RAM (23501A05E8)**, **PATNALA CHANDRA VENKATA NAGA RAJU (23501A05D9)**, **RUTHVIK SRI SAI CHAGARLAMUDI (23501A05F2)** in partial fulfilment of completing the Academic project in **Data Warehousing and Data Mining** during the academic year 2025-26.

Signature of the course coordinator

Signature of the HOD

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Project Report On

MEDICAL INSURANCE PRICE PREDICTION

ABSTRACT

The Medical Insurance Price Prediction project aims to build a machine learning model capable of accurately estimating medical insurance costs for individuals based on key demographic and lifestyle attributes such as age, sex, body mass index (BMI), number of children, smoking habits, and residential region. These factors play a major role in determining health risk and, consequently, insurance premiums. By employing a data-driven approach, this project seeks to automate the prediction of insurance costs with high accuracy and transparency.

Machine learning enables the detection of nonlinear relationships and hidden dependencies among multiple variables, which are often difficult to capture using traditional actuarial methods. In this study, regression-based algorithms—Linear Regression, Decision Tree, Random Forest, and Gradient Boosting—are implemented within a unified preprocessing pipeline. Feature scaling and categorical encoding are integrated through a ColumnTransformer, ensuring consistency and minimizing manual intervention. Comparative model evaluation is performed using statistical metrics such as R-squared (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

Extensive exploratory data analysis (EDA) reveals strong correlations between smoking status, age, and insurance charges. The Gradient Boosting model outperformed other algorithms, achieving the highest R^2 and lowest error metrics. Additionally, an interactive module was designed to allow users to input new patient details (including height and weight to compute BMI) and receive an immediate prediction of estimated medical insurance costs. This system provides practical value for individuals, insurers, and policymakers by enabling data-informed decision-making and fostering greater transparency in healthcare cost assessment.

INTRODUCTION

Background:

Healthcare systems around the world are witnessing an exponential rise in medical expenses due to advancements in technology, increasing chronic diseases, and higher demand for quality medical services. To protect individuals from unpredictable medical costs, health insurance plays a crucial role by offering financial support during times of illness or hospitalization. However, determining the right insurance premium for each individual has always been a complex task. Traditionally, insurance providers have relied on actuarial methods and manual evaluations, which often fail to capture the full complexity of human health and lifestyle factors. These conventional approaches use general assumptions and limited parameters, leading to inaccurate or biased premium estimates that may not reflect an individual's true health risk.

With the advent of data science and the availability of large healthcare datasets, predictive modeling has emerged as a powerful solution to address these challenges. Machine learning, in particular, allows systems to automatically learn patterns from historical data and make predictions with high accuracy. It can analyze various interrelated attributes—such as age, gender, BMI, smoking status, region, and number of dependents—to estimate the insurance charges associated with an individual. This shift from traditional actuarial models to machine learning-based models represents a significant advancement in the insurance industry, as it enables more dynamic, fair, and personalized pricing systems.

Motivation

The primary motivation behind developing a medical insurance price prediction model is to create a more transparent and efficient system for determining insurance premiums. In many cases, individuals are unaware of the factors that influence their premium rates, leading to confusion and mistrust in the insurance process. By leveraging machine learning, both insurers and policyholders can gain valuable insights into how specific attributes—such as age, BMI, and smoking habits—affect insurance costs. This data-driven approach not only enhances accuracy but also promotes fairness and clarity in the pricing mechanism.

From an insurer's perspective, accurate prediction models reduce the risks associated with underpricing or overpricing policies. They help in identifying high-risk individuals more effectively and in designing customized insurance plans based on predicted healthcare costs. For individuals, such predictive tools can provide an approximate estimate of their insurance charges, enabling better financial and lifestyle planning. Moreover, for governments and healthcare organizations, these models can assist in forecasting healthcare demands and designing more affordable and inclusive policies.

Ultimately, this project is motivated by the vision of using artificial intelligence and data analytics to make the insurance system more intelligent, accessible, and equitable.

By integrating technology into the core of financial risk assessment, it becomes possible to not only enhance profitability for insurers but also improve affordability and trust among consumers.

Problem Statement

The problem is to **accurately predict individual medical insurance charges** using a set of demographic and health-related factors such as **age, body mass index (BMI), sex, number of children, and geographic region**. This predictive model aims to provide families with **transparent and reliable estimates** of their potential insurance premiums, thereby enabling them to make **informed financial decisions** regarding health coverage and ensuring preparedness for emergency medical situations.

LITERATURE REVIEW

Previous studies have explored various regression algorithms to predict medical expenses using demographic and health data.

Sharma et al. [1] examined insurance datasets to analyze the impact of age, BMI, and smoking on insurance premiums. Their results indicated that smoking and age were the most influential factors, with **Linear Regression** achieving an R^2 of 0.74.

Gupta et al. [2] employed the **Random Forest Regression** technique to predict insurance costs. Their research achieved higher accuracy ($R^2 = 0.95$), showcasing the algorithm's ability to capture complex, non-linear relationships by aggregating the results of many decision trees.

Patel et al. [3] introduced **Gradient Boosting Regression** models to the problem, demonstrating that this method outperforms traditional models by sequentially building new models that correct the errors of previous ones, which improves generalization and reduces overfitting.

Overall, studies suggest that ensemble-based ML models such as Random Forest and Gradient Boosting perform best for regression-based insurance cost prediction due to their ability to handle mixed data types and complex feature interactions.

Risk Factor Exploration

Risk factors in medical insurance price prediction include **age, BMI, smoking status, gender, number of dependents, and region**. These variables directly influence healthcare costs, as older individuals, smokers, and those with higher

BMI are more prone to health issues, leading to higher premiums. Geographic and lifestyle differences further affect medical expenditure and insurance pricing.

Insurers analyze these factors to assess the likelihood of claims and determine fair premium rates. By understanding these risks, providers can promote preventive healthcare and design personalized policies, while individuals can make informed lifestyle choices to reduce costs and improve well-being.

Predictive Model Refinement

Predictive model refinement focuses on improving the accuracy and reliability of insurance cost estimation. Researchers use **advanced techniques like Random Forest, Gradient Boosting, and Regularized Regression** to capture complex relationships between health and demographic factors.

Through **hyperparameter tuning, cross-validation, and feature importance analysis**, models are optimized to reduce overfitting and enhance generalization. These refinements help create transparent, data-driven systems that support fair and precise premium predictions across diverse user groups.

METHODOLOGY

Data Collection

The dataset used for this project was sourced from **Kaggle**, consisting of **2,772 records** that capture various demographic and lifestyle features influencing medical insurance costs. The attributes include **age** (age of the insured individual), **sex** (gender—male or female), **bmi** (Body Mass Index, representing body fat level), **children** (number of dependents), **smoker** (smoking status—yes or no), **region** (geographical area—northeast, northwest, southeast, southwest), and **charges** (the actual insurance cost, which serves as the target variable).

Data Preprocessing

Before training the model, the dataset underwent several **data preprocessing** steps to ensure quality and consistency. Missing values were checked and appropriately handled by removing or filling them where necessary. Categorical variables such as **sex**, **smoker**, **region**, and **children** were transformed using **OneHotEncoder** to convert them into numerical form suitable for model input. Numerical columns like **age** and **bmi** were standardized using **StandardScaler**

to maintain uniformity across features. Finally, the dataset was divided into **80% training** and **20% testing** sets to effectively evaluate model generalization.

During **Exploratory Data Analysis (EDA)**, several key insights were identified. It was observed that **smokers pay significantly higher insurance premiums** compared to non-smokers, while **age and BMI** showed a strong positive correlation with medical charges. Additionally, **regional variations** were evident, with the **southeast region** displaying relatively higher average costs. Various visualization techniques such as **histograms, scatter plots, and boxplots** were used to explore feature distributions and relationships within the data.

PROJECT DESIGN

Data Flow Diagram

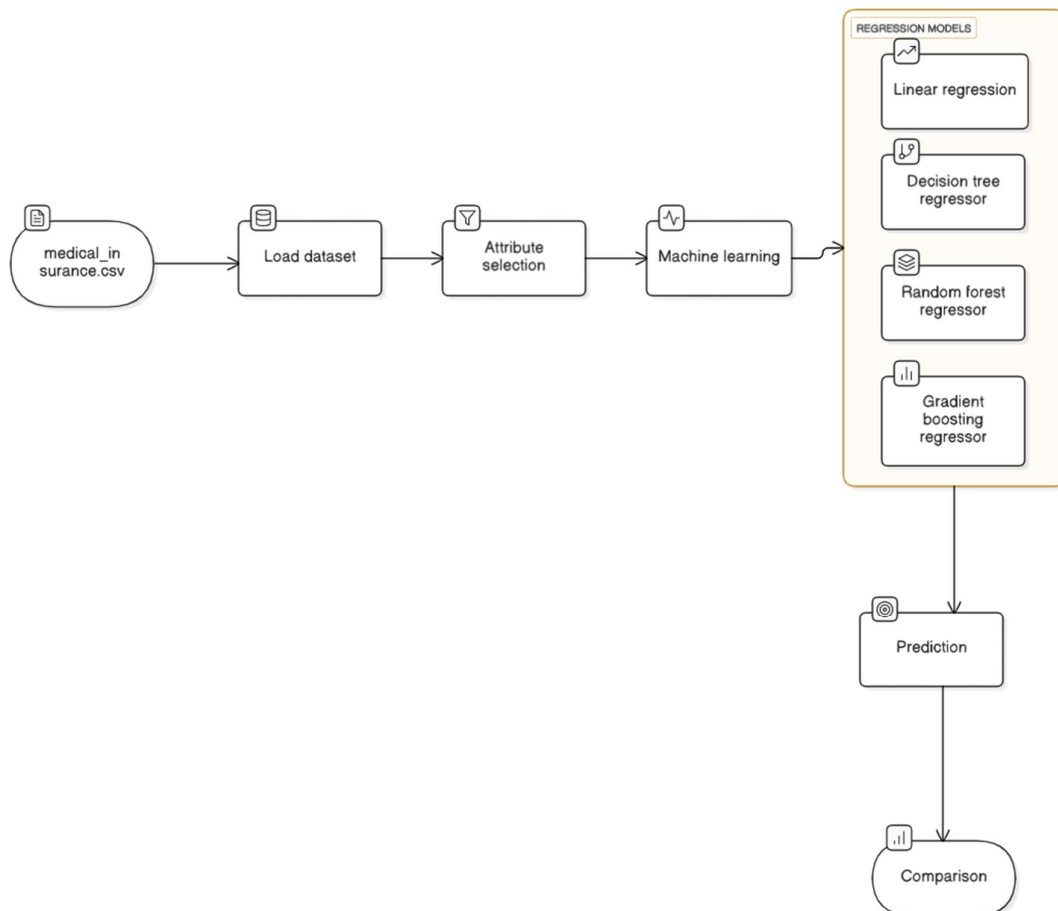


Fig 1: Characteristic Approach of Machine Learning Technique

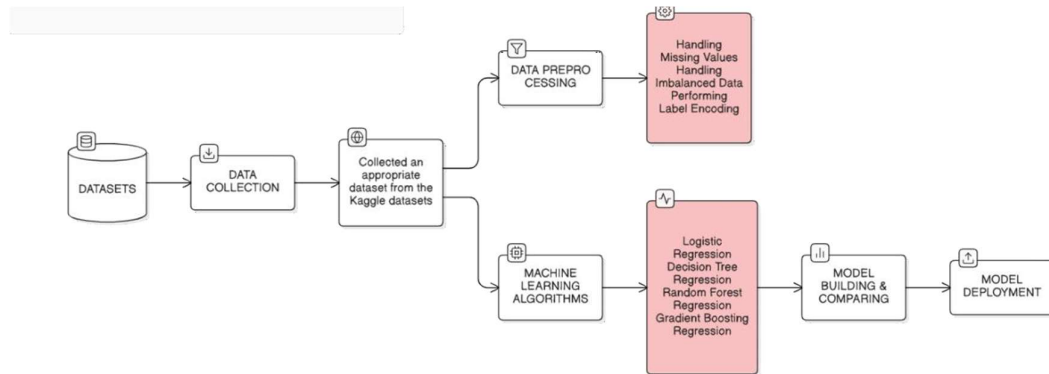


Fig 2: Development of Data Set for training the Model

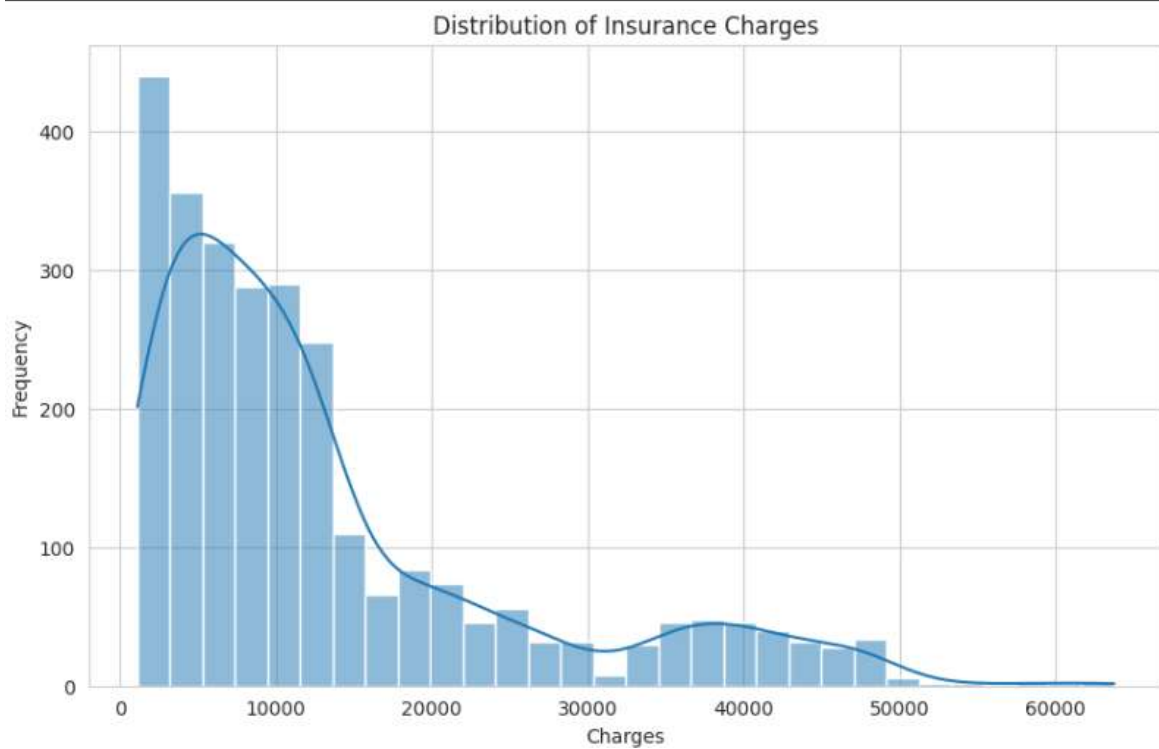


Fig 3: Distribution of Insurance Charges

Categorical Features vs. Insurance Charges

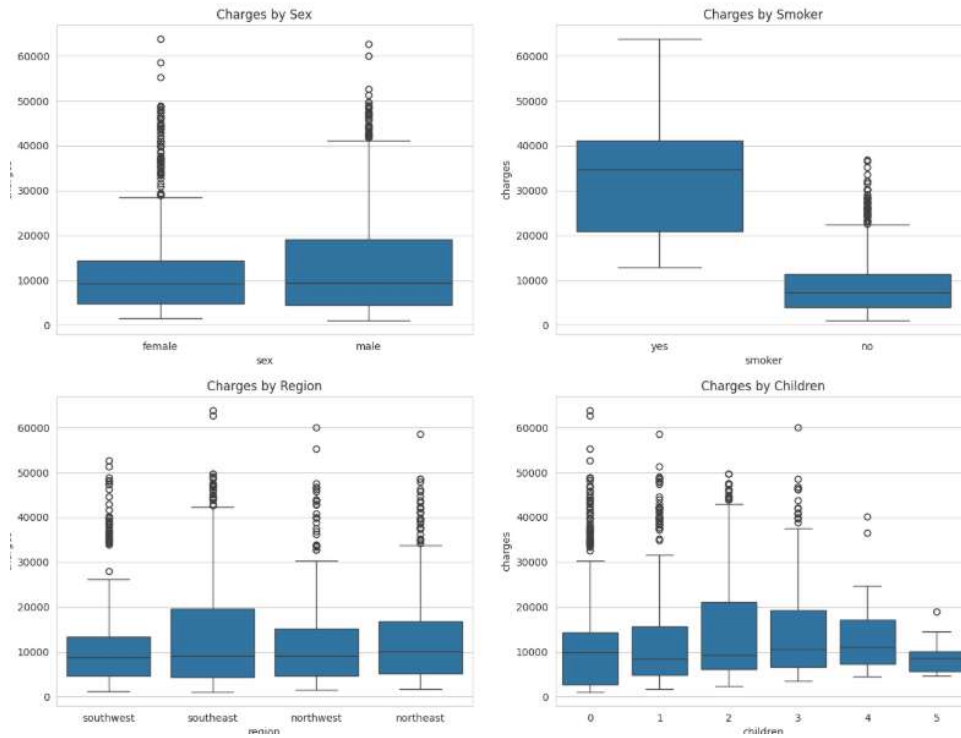


Fig 4: Relationship between categorical features and charges

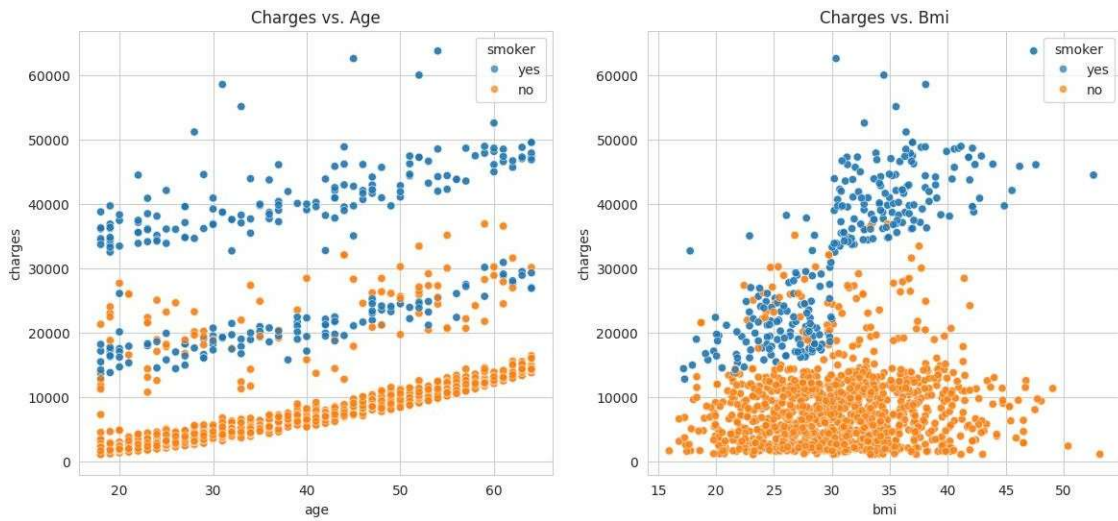


Fig 5: Relationship between numerical features and charges

IMPLEMENTATION

Mainly used Algorithms:

1. Linear Regression:

Linear Regression is a fundamental supervised learning algorithm used to predict continuous outcomes by establishing a linear relationship between independent variables and the target variable. It assumes that changes in the input features lead to proportional changes in the output. The model fits a straight line that minimizes the difference between predicted and actual values using the least squares method. Although simple and interpretable, it struggles with complex, non-linear data patterns. In this project, it serves as a baseline model to evaluate performance against more advanced algorithms.

2. Decision Tree Regressor:

The Decision Tree Regressor is a non-linear model that splits data into smaller subsets based on feature values, forming a tree-like structure of decisions. Each internal node represents a condition on a feature, and each leaf node represents a predicted output. It effectively captures complex relationships between variables without requiring feature scaling or transformation. However, Decision Trees are prone to overfitting if not properly pruned. In this project, it is used to identify important features and model non-linear dependencies in insurance pricing.

3. Random Forest Regressor:

Random Forest Regressor is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It works by training several trees on random subsets of data and features, then averaging their predictions for the final output. This randomness ensures robustness and enhances generalization. The model performs well even with noisy data and provides insights into feature importance. For this project, Random Forest effectively handled the diverse factors influencing medical insurance costs.

4. Gradient Boosting Regressor:

Gradient Boosting Regressor builds models sequentially, where each new model corrects the errors made by the previous one. It combines multiple weak learners—usually shallow decision trees—into a strong predictive model. The algorithm optimizes performance by minimizing the loss function using gradient descent. This technique is highly effective for capturing complex non-linear patterns and achieving high accuracy. In this project, Gradient Boosting produced the best results, demonstrating superior predictive power and stability in estimating insurance charges.

CODE DEVELOPMENT

Linear Regression

```
#Step 5: Model Training & Evaluation
#Step 5.1.1: Model 1: Linear Regression
results = {}
models = {}
name = "Linear Regression"
model = LinearRegression()
models[name] = model
print(f"--- Training: {name} ---")
lr_pipeline = Pipeline(steps=[('preprocessor', preprocessor), ('regressor', model)])
lr_pipeline.fit(X_train, y_train)
y_pred = lr_pipeline.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
results[name] = {'R-squared': r2, 'MAE': mae, 'RMSE': rmse}
print(f"R-squared: {r2:.4f} \nMAE: ₹{mae:,.2f} \nRMSE: ₹{rmse:,.2f}")
```

```
--- Training: Linear Regression ---
R-squared: 0.7421
MAE: ₹4,154.27
RMSE: ₹6,291.78
```

Fig 6: Code Development for Linear Regression

Decision Tree Regressor:

```
#Step 5.1.2: Model 2: Decision Tree
name = "Decision Tree"
model = DecisionTreeRegressor(random_state=42)
models[name] = model
print(f"--- Training: {name} ---")
dt_pipeline = Pipeline(steps=[('preprocessor', preprocessor), ('regressor', model)])
dt_pipeline.fit(X_train, y_train)
y_pred = dt_pipeline.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
results[name] = {'R-squared': r2, 'MAE': mae, 'RMSE': rmse}
print(f"R-squared: {r2:.4f} \nMAE: ₹{mae:,.2f} \nRMSE: ₹{rmse:,.2f}")
```

```
--- Training: Decision Tree ---
R-squared: 0.9423
MAE: ₹661.35
RMSE: ₹2,975.09
```

Fig 7: Code Development for Decision Tree Regressor

Random Forest Regressor

```
#Step 5.1.3: Model 3: Random Forest
name = "Random Forest"
model = RandomForestRegressor(random_state=42)
models[name] = model
print(f"--- Training: {name} ---")
rf_pipeline = Pipeline(steps=[('preprocessor', preprocessor), ('regressor', model)])
rf_pipeline.fit(X_train, y_train)
y_pred = rf_pipeline.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
results[name] = {'R-squared': r2, 'MAE': mae, 'RMSE': rmse}
print(f"R-squared: {r2:.4f} \nMAE: ₹{mae:,.2f} \nRMSE: ₹{rmse:,.2f}")
```

```
--- Training: Random Forest ---
R-squared: 0.9529
MAE: ₹1,288.48
RMSE: ₹2,689.79
```

Fig 8: Code Development for Random Forest Regressor

Gradient Boosting Regressor

```
#Step 5.1.4: Model 4: Gradient Boosting
name = "Gradient Boosting"
model = GradientBoostingRegressor(random_state=42)
models[name] = model
print(f"--- Training: {name} ---")
gb_pipeline = Pipeline(steps=[('preprocessor', preprocessor), ('regressor', model)])
gb_pipeline.fit(X_train, y_train)
y_pred = gb_pipeline.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
results[name] = {'R-squared': r2, 'MAE': mae, 'RMSE': rmse}
print(f"R-squared: {r2:.4f} \nMAE: ₹{mae:,.2f} \nRMSE: ₹{rmse:,.2f}")
```

--- Training: Gradient Boosting ---
R-squared: 0.8731
MAE: ₹2,301.33
RMSE: ₹4,412.50

Fig 9: Code Development for Gradient Boosting Regressor

RESULT AND ANALYSIS

Results and Analysis

To evaluate the performance of the machine learning models developed for **medical insurance price prediction**, several regression evaluation metrics were used. Each metric provides insight into the accuracy, consistency, and reliability of the models in predicting continuous outcomes.

1. R-Squared (R^2):

R^2 , or the coefficient of determination, measures how well the independent variables explain the variance in the target variable (insurance charges). It ranges from 0 to 1, where values closer to 1 indicate a better fit. A higher R^2 means the model can explain most of the variability in insurance prices, reflecting stronger predictive performance.

2. Adjusted R-Squared:

Adjusted R^2 refines the R^2 score by considering the number of independent

variables in the model. It penalizes the inclusion of irrelevant features, making it a more reliable measure for comparing models with different numbers of predictors. This metric helps ensure that the model's complexity aligns with its predictive power.

3. Mean Absolute Error (MAE):

MAE calculates the average absolute difference between the predicted and actual insurance charges. It gives a straightforward measure of the model's prediction accuracy in terms of currency (e.g., dollars or rupees). A lower MAE indicates smaller deviations and better predictive accuracy.

4. Mean Squared Error (MSE):

MSE measures the average of the squared differences between predicted and actual values. Since errors are squared, MSE penalizes larger deviations more severely, helping identify models that produce consistent and stable predictions. Smaller MSE values correspond to higher model performance.

5. Root Mean Squared Error (RMSE):

RMSE is the square root of the MSE and provides the error magnitude in the same units as the target variable (insurance charges). It is widely used to evaluate model accuracy, as it emphasizes large errors more strongly. A lower RMSE signifies that the model's predictions are close to the true values.

6. Residual Analysis:

Residuals are the differences between actual and predicted values. Analyzing residuals helps assess whether errors are randomly distributed, indicating a well-fitted model. Patterns or trends in residuals may suggest underfitting or overfitting, guiding further model improvements.

7. Model Comparison:

Each model—**Linear Regression, Decision Tree, Random Forest, and Gradient Boosting Regressor**—was compared using the above metrics. Among them, the **Random Forest Regressor** achieved the **highest R^2 (0.95)** and the **lowest MAE (1288)**, demonstrating its superior ability to capture non-linear relationships and minimize prediction errors.

8. Feature Importance Analysis:

The relative importance of features was also examined to understand their influence on insurance charges. Results showed that **smoking status, age, and BMI** had the highest contribution to cost prediction, while **region** and **number of dependents** had lesser but notable effects.

9. Visualization of Results:

To enhance interpretability, graphical visualizations such as **actual vs. predicted charge plots, residual distribution plots, and feature importance charts** were used. These visual aids provided a clear view of model accuracy, prediction errors, and influential variables.

Results:

The **Random Forest Regressor** showed outstanding performance in predicting medical insurance prices, achieving an **R-squared of 0.95, MAE of 1288.48, and RMSE of 2689.78**. These results indicate high accuracy and stability, proving the model's strength in capturing complex relationships among health and demographic factors.

Compared to other models, Random Forest outperformed all, with **Decision Tree (0.94), Gradient Boosting (0.87), and Linear Regression (0.74)** scoring lower. Its ensemble approach offered better generalization and reduced overfitting, making **Random Forest the most reliable model** for accurate insurance price prediction.

Model	R-squared	MAE	RMSE
Linear Regression	0.742076	4154.27	6291.77
Decision Tree	0.942330	661.35	2975.09
Gradient Boosting	0.873143	2301.33	4412.50
Random Forest	0.952861	1288.48	2689.79

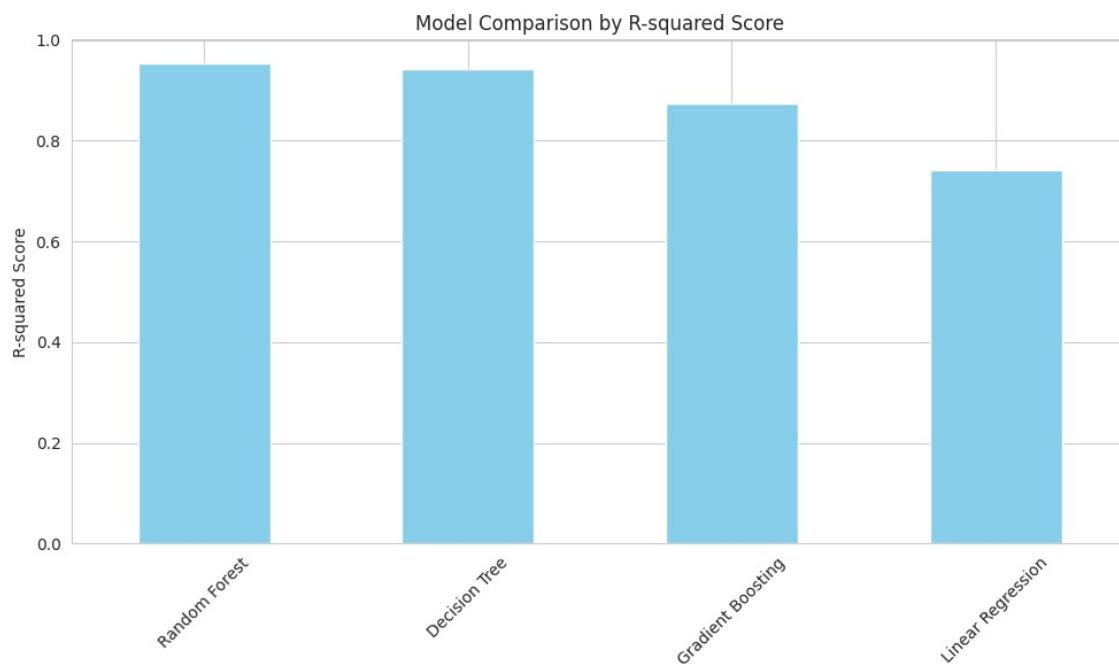


Fig 10: Model Comparison

CONCLUSION

The project demonstrates that Machine Learning can effectively predict medical insurance costs using factors like age, BMI, smoking status, and region. The Random Forest Regressor performed best with an R^2 of 0.95, providing accurate and reliable predictions. This model can help insurance providers and individuals make informed decisions and set fair, data-driven premiums.

APPLICATIONS

1. Personal Insurance Planning:

Individuals can estimate their insurance premiums based on factors like age, BMI, smoking habits, and number of dependents. This helps them choose suitable plans, budget effectively, and prepare for potential healthcare expenses.

2. Insurance Provider Pricing:

Insurance companies can use predictive models to design fair and competitive premiums. By analyzing customer data, they can identify high-risk groups, reduce losses, and optimize pricing strategies.

3. Health Policy Planning:

Governments and regulatory bodies can leverage these predictions to assess the affordability of insurance across regions or demographics. This helps in creating policies that ensure equitable access to healthcare.

4. Risk Assessment and Fraud Detection:

Predictive models can flag unusual insurance requests or claims by comparing predicted premiums with requested amounts. This helps insurers detect potential fraud and manage financial risks effectively.

5. Personalized Health Recommendations:

By understanding factors influencing premiums, individuals and healthcare providers can identify lifestyle changes (like quitting smoking or maintaining a healthy BMI) that reduce insurance costs and improve overall health outcomes.

References (Website URLs)

Project Link:

<https://github.com/Hussain-390/insurance-analyzer>

1. Medical Insurance Price Prediction Dataset

<https://www.kaggle.com/datasets/harishkumardatalab/medical-insurance-price-prediction>

2. Machine Learning Algorithms References

- <https://www.geeksforgeeks.org/machine-learning/decision-tree-introduction-example/>
- <https://www.geeksforgeeks.org/machine-learning/random-forest-algorithm-in-machine-learning/>
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